



INTRODUCTION TO SDLC

- ❖ SDLC Stands for **Software Development Life Cycle**.
- ❖ The software development life cycle (SDLC) is a framework defining tasks performed at each step in the software development process.
- ❖ It consists of a detailed plan describing how to develop, maintain and replace specific software.

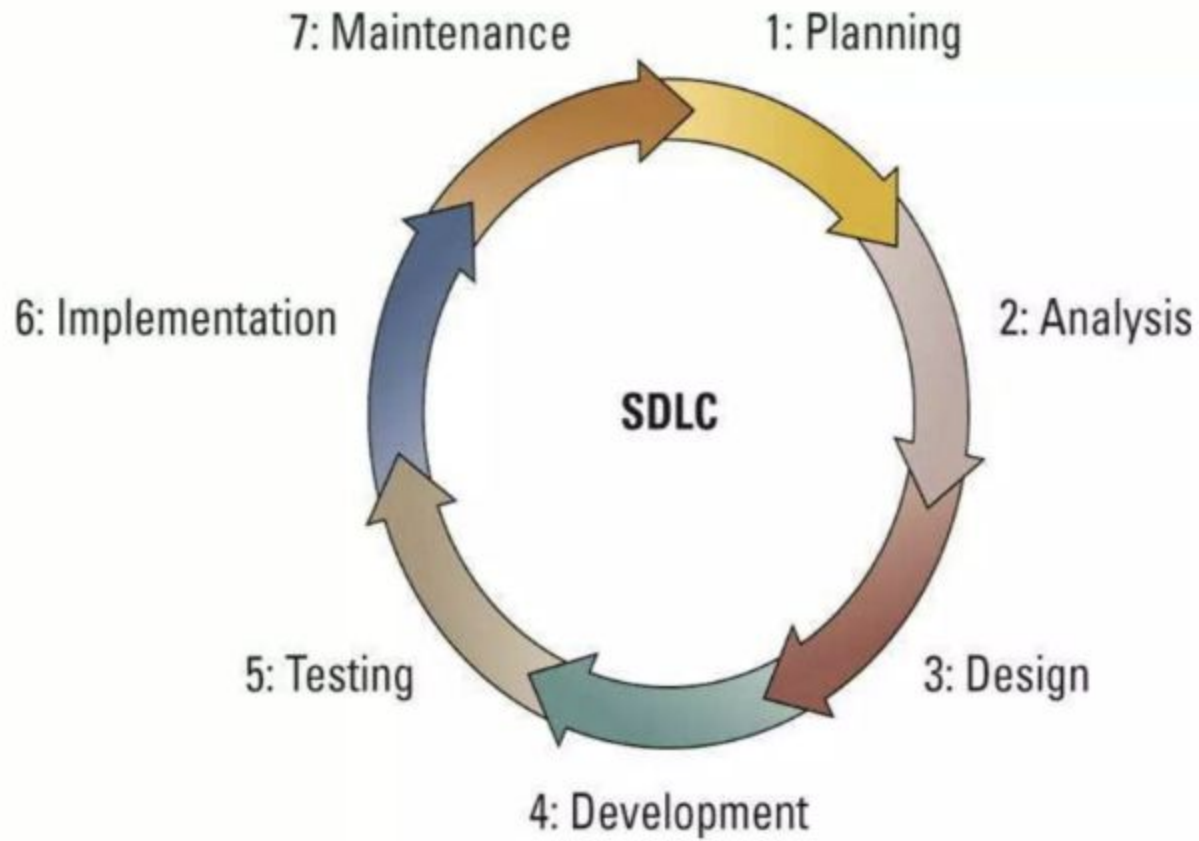


OBJECTIVE OF SDLC

- SDLC has Three primary objectives:
 - ✓ Ensure the delivery of high quality systems.
 - ✓ Provide strong management control.
 - ✓ Maximize productivity.



SDLC PHASES



Phase 1: Planning

- Determine if a new system is needed
- Three primary tasks:
 - ✓ Define the problem.
 - ✓ Suggest alternative solutions.
 - ✓ Prepare a short report.



Phase 2: System Analysis

- In depth study of the existing system to determine what the new system should do.
- In addition to observation and interviews, examine:
 - ✓ Formal lines of authority (org chart)
 - ✓ Standard operating procedures
 - ✓ How information flows
 - ✓ Reasons for any inefficiencies



Phase 3: System Design

- It takes the initial input
- For each requirements design elements will be produced
- It **describes** the software features and includes **hierarchy diagrams , screen layout diagrams**
- The **output** of the stage describe the new system as a collection of modules or subsystems.



Phase 4: System Development

- Build the system to the design specifications
 - ✓ Develop the software
 - ✓ Acquire the hardware
- Create manuals for users and operators



PHASE 5: SYSTEM TESTING

- Test the new system
 - ✓ Module (unit) test - tests each part of system
 - ✓ Integration testing - tests system as one unit



Phase 6: System Implementation

- ✓ Convert from old system to new system.
- ✓ Train users
- ✓ Compile final documentation
- ✓ Evaluate the new system
- ✓ User Training



SDLC MODELS

➤ A framework that describes the activities performed at each stage of a software development project.

- ✓ Waterfall model
- ✓ Iterative model
- ✓ Prototyping model
- ✓ Spiral Model
- ✓ V model



Phase 7: System Maintenance

- ✓ Physical repair of the system.
- ✓ Correction of new bugs found.
- ✓ System adjustments to environmental changes.
- ✓ Adjustments for users' changing needs (adaptive).
- ✓ Changes to user better techniques when they become available (perfective).

